

Experiment record: context-composed TabPFN for long-tailed NetFlow intrusion detection

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1 Claim and research question

A frozen tabular foundation model (TabPFN-v3) adapts to newly appearing attack traffic by changing its context rows rather than its weights. On the same 100k training rows its macro-F1 matches XGBoost (Table 1); the research question is what to put into a context budget that is far smaller than the available training data so that tail attacks are recovered, and how adaptation cost (labels, time) compares with retraining. The target is tail recovery and benign false positives, not all-class macro-F1.

2 Data and protocol

NetFlow-v3 CIC-IDS2018 and ToN-IoT, chronological splits per attack scenario, exact-duplicate label conflicts removed (2026-09-15/16 fixed splits). Test rows: CIC-IDS2018 3,042,473; ToN-IoT 2,271,723. All numbers are seed 43 unless stated; test is a previously observed development holdout, so no result below is a final claim.

3 Baselines on the cleaned data (EXP61)

Table 1: Macro-F1 and accuracy on the conflict-free test splits (seed 43). The first five methods use exactly the same 100,000 training IDs as the Global context C0; full-train XGBoost and the proposed model use additional training data and are not same-budget comparisons.

Method	CIC macro-F1	CIC acc. (%)	ToN macro-F1	ToN acc. (%)	Training rows (CIC / ToN)
Global TabPFN-v3 (C0)	0.7824	99.37	0.6796	90.13	100,000
XGBoost (same 100k)	0.7821	98.83	0.6983	92.09	100,000
BoostPFN	0.7148	98.90	0.5866	82.68	100,000
LoCalPFN (FT)	0.7024	97.72	0.6462	89.44	100,000
DistPFN (v3)	0.7839	99.43	0.6682	88.56	100,000
XGBoost (full train)	0.7795	98.02	0.7162	92.38	8,666,430 / 4,669,119
Proposed: experts (K=6) + S/V	0.7824	99.37	0.7677	90.68	100,000 + expert/route pools

Bold: column maximum. Test rows: CIC-IDS2018 3,042,473; ToN-IoT 2,271,723. Source: EXP61 (SOTA), EXP63 K sweep (proposed, K=6: the best ToN macro-F1 of K in 2, 4, 6, 8, chosen on test, single seed).

Table 2: Measured cost on one RTX 4090 (24 GB) with up to two jobs in parallel: elapsed times include resource contention and are not standalone latencies. LoCalPFN fit includes validation; its prediction includes kNN retrieval. DistPFN shares the Global fit and prediction.

Method	CIC fit (s)	CIC predict (s)	ToN fit (s)	ToN predict (s)	Peak GPU GiB (CIC / ToN)
Global TabPFN-v3 (C0)	36.4	1,099.3	36.7	821.7	2.92 / 2.94
XGBoost (same 100k)	2.3	1.0	55.4	1.0	0.55 / 0.53
BoostPFN	98.5	2,308.5	97.8	1,737.6	0.89 / 0.89
LoCalPFN (FT)	3,196.3	17,300.8	5,783.1	11,144.8	7.54 / 7.54
DistPFN (v3)	36.4	1,099.4	36.7	821.9	2.92 / 2.94
XGBoost (full train)	107.8	1.0	143.2	1.1	2.20 / 1.59

Table 3: Per-class F1 on CIC-IDS2018 (seed 43, test rows in descending order). Proposed = experts (K=6) + scorer/verifier (S1V0 policy), EXP63 K sweep; bold marks the row maximum.

Class	Test rows	Global TabPFN-v3	XGBoost (100k)	BoostPFN	LoCalPFN	DistPFN	XGBoost (full)	Proposed K=6
Benign	2,652,203	0.9965	0.9933	0.9937	0.9921	0.9968	0.9887	0.9965
DDoS	262,680	0.9922	0.9998	0.9906	0.9433	0.9931	0.9036	0.9922
Bot	41,542	0.9985	0.9976	0.9942	0.9972	0.9982	0.9977	0.9985
DoS	39,483	0.9920	0.9998	0.9598	0.7309	0.9919	0.9999	0.9920
Brute force	37,696	0.9997	0.9995	0.9997	0.9999	0.9997	0.9996	0.9997
Infiltration	8,742	0.2879	0.1528	0.0653	0.1350	0.3055	0.2927	0.2880
Web attack	127	0.2102	0.3321	0.0000	0.1183	0.2024	0.2745	0.2102

Table 4: Per-class F1 on ToN-IoT (seed 43, test rows in descending order). Proposed = experts (K=6) + scorer/verifier (S1V0 policy), EXP63 K sweep; bold marks the row maximum.

Class	Test rows	Global TabPFN-v3	XGBoost (100k)	BoostPFN	LoCalPFN	DistPFN	XGBoost (full)	Proposed K=6
Benign	1,474,302	0.9217	0.9385	0.9042	0.9184	0.9064	0.9389	0.9258
XSS	343,854	0.8305	0.8734	0.7929	0.8469	0.8247	0.8714	0.8408
Password	210,951	0.8921	0.8920	0.6728	0.8911	0.8909	0.8918	0.8922
DDoS	103,338	0.9804	0.9842	0.9248	0.9786	0.9850	0.9928	0.9865
Injection	68,891	0.9027	0.9128	0.4793	0.8925	0.9164	0.9759	0.9041
Backdoor	40,030	0.9990	0.9990	0.9609	0.9988	0.9990	0.9924	0.9990
DoS	22,458	0.9771	0.9563	0.3954	0.5554	0.9823	0.9814	0.9769
Scanning	6,029	0.0394	0.0458	0.3281	0.0973	0.0060	0.0396	0.8995
MitM	1,161	0.1322	0.2363	0.0468	0.1655	0.0512	0.3236	0.1321
Ransomware	709	0.1208	0.1449	0.3608	0.1170	0.1204	0.1539	0.1201

4 Residual experts and scorer/verifier (EXP59, EXP62, EXP63)

Table 5: Label-assisted residual-assignment oracle (EXP59, K=4): each test row is assigned to the expert of its nearest training residual cluster (the residual uses the test label) and that expert's prediction is scored as is. It is a diagnostic, not an attainable bound; for the Benign/Infiltration pair the assignment leaks the label, and for MitM the oracle is worse than Global.

Dataset	Class	Test rows	G P	G R	G F1	Oracle P	Oracle R	Oracle F1
CIC-IDS2018	Benign	2,652,203	0.9961	0.9969	0.9965	0.9997	0.9999	0.9998
CIC-IDS2018	DDoS	262,680	1.0000	0.9845	0.9922	1.0000	0.9993	0.9997
CIC-IDS2018	Bot	41,542	0.9999	0.9970	0.9985	0.9999	0.9971	0.9985
CIC-IDS2018	DoS	39,483	0.9997	0.9844	0.9920	0.9999	0.9930	0.9965
CIC-IDS2018	Brute force	37,696	1.0000	0.9995	0.9997	1.0000	0.9995	0.9997
CIC-IDS2018	Infiltration	8,742	0.2612	0.3209	0.2880	0.9666	0.9345	0.9503
CIC-IDS2018	Web attack	127	0.1185	0.9291	0.2102	0.2293	0.7638	0.3527
ToN-IoT	Benign	1,474,302	0.9765	0.8728	0.9217	0.9989	0.8344	0.9093
ToN-IoT	XSS	343,854	0.7490	0.9318	0.8305	0.7431	0.9975	0.8517
ToN-IoT	Password	210,951	0.8072	0.9969	0.8921	0.8070	0.9976	0.8922
ToN-IoT	DDoS	103,338	0.9896	0.9713	0.9804	0.9879	0.9807	0.9843
ToN-IoT	Injection	68,891	0.8565	0.9543	0.9027	0.9204	0.9862	0.9522
ToN-IoT	Backdoor	40,030	0.9985	0.9996	0.9990	0.9983	0.9998	0.9990
ToN-IoT	DoS	22,458	0.9707	0.9835	0.9771	0.9594	0.9955	0.9771
ToN-IoT	Scanning	6,029	0.7722	0.0202	0.0394	0.5916	0.9174	0.7193
ToN-IoT	MitM	1,161	0.0713	0.8984	0.1322	0.0200	0.9802	0.0393
ToN-IoT	Ransomware	709	0.0647	0.9154	0.1208	0.0594	0.9958	0.1121

Table 6: Scorer/verifier (S1V0) on the seed-43 banks with K = 2, 4, 6, 8 (EXP62/63). Calls and acceptances are logical counts obtained by applying the validation-selected policy to stored expert predictions. CIC-IDS2018 keeps the Global prediction at every K; ToN-IoT calls one expert per row and the gain is almost entirely Scanning.

Dataset	Bank	Macro-F1	Call rate (%)	Accept rate (%)	Changed	Corrected	Damaged
CIC-IDS2018	Global	0.7824	0.00	0.00	0	0	0
CIC-IDS2018	K = 2	0.7824	0.00	0.00	0	0	0
CIC-IDS2018	K = 4	0.7824	0.00	0.00	0	0	0
CIC-IDS2018	K = 6	0.7824	0.00	0.00	0	0	0
CIC-IDS2018	K = 8	0.7824	0.00	0.00	0	0	0
ToN-IoT	Global	0.6796	0.00	0.00	0	0	0
ToN-IoT	K = 2	0.7636	100.00	5.26	17,201	10,443	5,784
ToN-IoT	K = 4	0.7617	100.00	1.23	6,504	5,381	1,101
ToN-IoT	K = 6	0.7677	100.00	4.09	18,447	15,289	2,771
ToN-IoT	K = 8	0.7463	100.00	0.69	12,290	8,328	3,741

5 Where the benign confusion comes from (2026-10-01 diagnosis)

Nearest-neighbour distances of Global's errors to the training pool (standardised 46-dimensional L2; a typical benign row is 0.01 from training benign): Scanning misses lie 0.60 from training Scanning and 2.92 from benign (a context prior problem, fixed by an expert with a Scanning block plus the verifier: F1 0.039 to 0.90); Infiltration false positives lie 0.11 / 0.17 from both (overlapping region, a feature

limit shared by every method); Ransomware false positives lie 0.12 from training Ransomware and 2.84 from benign (near-duplicate label contradictions); MitM false positives lie about 10 from every training class (a benign block that only exists in the test period, i.e. drift).

6 Context composition feasibility (EXP64–67, 2026-10-01)

Table 7: Feasibility of the four follow-up approaches (2026-10-01, seed 43, single runs). Subsample = stratified screening subset with 20k rows per class; it inflates tail precision and is used only for relative comparison.

Approach	Change	Result (full test unless noted)	Verdict
1 Benign diversity (ToN)	C0 benign rows re-selected (time / k-center / inverse density / hard)	0.830 $\rightarrow$ 0.813–0.834 (subsample)	no effect
2 LITO-style synthesis (CIC web)	224 real rows $\rightarrow$ 2,000 synthetic, 1479 self-authenticated	macro 0.7824 $\rightarrow$ synthetic 0.7680; real duplicates 0.7997 (web F1 0.210 $\rightarrow$ 0.320)	synthesis rejected; tail ratio is a knob
3 SupCon prototype embedding (ToN)	16-d encoder trained on the route pool, appended to features	0.836 $\rightarrow$ 0.827 (raw+emb), 0.836 (emb) (subsample)	no effect
4 Context optimisation (ICD / prompt tuning)	context features optimised by gradient through the frozen TabPFN, natural-ratio train queries, 300 steps	ToN 3,000-row context: 0.7100 / 0.7455 (seeds 43/44); CIC 2,100-row: 0.7829; balanced-query objective 0.6720	exceeds the 100k random C0 (ToN 0.6796, CIC 0.7824)

7 Temporal-arrival protocol (EXP68)

Table 8: Temporal-arrival protocol on ToN-IoT (EXP68, seed 43): test rows in time order, five equal periods; in each period the first 10% of every family's rows may be labelled (at most 50 per class) and the rest is evaluated. Macro-F1 over the classes present in the period. TabPFN + context appends the labelled rows to C0 without any weight update; XGB retrained refits on C0 plus the accumulated labels.

Period	Attack families present	Eval rows	TabPFN frozen	TabPFN + context	XGB frozen	XGB retrained	Labels
1	DDoS, DoS, Injection, Scanning	408,908	0.7929	0.7964	0.7900	0.7986	250
2	DDoS	408,910	0.9724	0.9758	0.9763	0.9764	350
3	Password	408,909	0.8677	0.8676	0.8744	0.8740	450
4	XSS	408,909	0.9060	0.9415	0.9607	0.9613	550
5	Backdoor, MitM, Ransomware	408,908	0.5628	0.5540	0.6050	0.6096	750

8 Decision log (branch points)

- 2026-09-15: data-audit experiments (EXP40–44) closed; contradictions are reported as per-class ceilings, not removed from the headline benchmark; focus moved to improving the context-composed model.
- 2026-09-22: residual oracle redefined (label-assisted assignment, designated expert scored as is); union-of-experts oracles are no longer cited as expert capability.
- 2026-09-29: class-level specialist substitution without row routing and the open-set leg were rejected by the author; the model structure (global $\rightarrow$ scorer $\rightarrow$ top-1 expert $\rightarrow$ verifier) is fixed.

- 2026-09-30: EXP60 (random vs. nearby counterexample contexts) withdrawn from the evidence base; EXP61 SOTA on cleaned data and EXP62/63 S/V completed; three report tabs merged.
- 2026-10-01: the earlier A/B interpretation ("experts are good in their own region but cannot separate similar classes") was retracted in favour of direct same-population evaluation; four follow-up approaches screened, only context optimisation produced new information.
- 2026-10-02: the deck and the protocol were reframed claim-first: adaptation without retraining, tail recovery as the target, temporal-arrival evaluation.